Lab 6: Code and Branch Coverage Analysis

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Repository: L6_L7_51_self_storage

Module under test: self_storage.py

1. Module overview

self_storage.py has three functions: rent_due (rent after deposit months, member bonus month, climate rate and long stay discount), access_allowed (owner and visitor access rules by hour and visitor count) and payment_status (current, grace or lien based on days to due date, autopay and prior late payments).

2. Commands used

pip install coverage pytest

coverage run -m pytest <test_file> then coverage report -m (line coverage)

coverage run --branch -m pytest <test_file> then coverage report -m (branch coverage)

3. Baseline (starter test file)

The starter test_self_storage.py has 2 tests. Coverage of self_storage.py:

Metric	Result	Missing
Line coverage	38% (23 of 37 statements missed)	6, 8, 11, 15-23, 28, 30, 35-45
Branch coverage	32% (6 partial branches)	same lines, most branches never taken

4. Code coverage test cases

CodeCoverageTest_self_storage.py has 11 tests chosen so every line runs at least once: invalid unit size, negative months, rent inside deposit months, short stay without climate, long stay member with climate, owner access, access outside hours, normal access, and current, grace and lien payment states.

Metric	Result
Tests passed	11 of 11
Line coverage of self_storage.py	100%
Branch coverage of self_storage.py	98% (1 partial branch: 21->23)

Observation: line coverage is 100% but branch coverage is not. Line 21 (if is_climate inside the long stay discount) only ever evaluated to True, so the path where a long stay has no climate control was never tested. This shows that full line coverage does not mean every decision was tested.

5. Branch coverage test cases

BranchCoverageTest_self_storage.py has 28 tests. On top of the code coverage cases it makes every decision go both ways and checks boundaries: months equal to 0, extra months equal to 0 and exactly 6, member on a large unit (no bonus month), long stay without climate, owner with too many visitors, hours 5, 6, 22 and 23, too many visitors, all three payment states for autopay and manual pay, and prior late payment with and without autopay.

Metric	Result
Tests passed	28 of 28
Line coverage of self_storage.py	100%

Branch coverage of self_storage.py	100% (26 of 26 branches, 0 partial)

6. Summary

Test file	Tests	Line coverage	Branch coverage
test_self_storage.py (starter)	2	38%	32%
CodeCoverageTest_self_storage.py	11	100%	98%
BranchCoverageTest_self_storage.py	28	100%	100%

7. Screenshots

Code coverage tests - line coverage

```
$ coverage run -m pytest CodeCoverageTest_self_storage.py
============================= test session starts ==============================
platform linux -- Python 3.13.18, pytest-9.1.1, pluggy-1.6.0
rootdir: /work/L6_L7_51_self_storage/Lab 6
plugins: platformdirs-4.13.0, anyio-4.15.1
collected 11 items

CodeCoverageTest_self_storage.py ...........                             [100%]

============================== 11 passed in 0.04s ==============================
$ coverage report -m
Name                              Stmts   Miss  Cover   Missing
---------------------------------------------------------------
CodeCoverageTest_self_storage.py     26      0   100%
self_storage.py                      37      0   100%
---------------------------------------------------------------
TOTAL                                63      0   100%
```

Code coverage tests - branch coverage

```
$ coverage run --branch -m pytest CodeCoverageTest_self_storage.py
============================= test session starts ==============================
platform linux -- Python 3.13.18, pytest-9.1.1, pluggy-1.6.0
rootdir: /work/L6_L7_51_self_storage/Lab 6
plugins: platformdirs-4.13.0, anyio-4.15.1
collected 11 items

CodeCoverageTest_self_storage.py ...........                             [100%]

============================== 11 passed in 0.03s ==============================
$ coverage report -m
Name                              Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------------------
CodeCoverageTest_self_storage.py     26      0      0      0   100%
self_storage.py                      37      0     26      1    98%   21->23
-----------------------------------------------------------------------------
TOTAL                                63      0     26      1    99%
```

Branch coverage tests - line coverage

```
$ coverage run -m pytest BranchCoverageTest_self_storage.py
============================= test session starts ==============================
platform linux -- Python 3.13.18, pytest-9.1.1, pluggy-1.6.0
rootdir: /work/L6_L7_51_self_storage/Lab 6
plugins: platformdirs-4.13.0, anyio-4.15.1
collected 28 items

BranchCoverageTest_self_storage.py ............................          [100%]

============================== 28 passed in 0.10s ==============================
$ coverage report -m
Name                                 Stmts   Miss  Cover   Missing
------------------------------------------------------------------
BranchCoverageTest_self_storage.py      64      0   100%
self_storage.py                         37      0   100%
------------------------------------------------------------------
TOTAL                                  101      0   100%
```

Branch coverage tests - branch coverage

```
$ coverage run --branch -m pytest BranchCoverageTest_self_storage.py
============================= test session starts ==============================
platform linux -- Python 3.13.18, pytest-9.1.1, pluggy-1.6.0
rootdir: /work/L6_L7_51_self_storage/Lab 6
plugins: platformdirs-4.13.0, anyio-4.15.1
collected 28 items

BranchCoverageTest_self_storage.py ............................          [100%]

============================== 28 passed in 0.04s ==============================
$ coverage report -m
Name                                 Stmts   Miss Branch BrPart  Cover   Missing
--------------------------------------------------------------------------------
BranchCoverageTest_self_storage.py      64      0      0      0   100%
self_storage.py                         37      0     26      0   100%
--------------------------------------------------------------------------------
TOTAL                                  101      0     26      0   100%
```